

Rayan Aslam

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EDUCATION

Rutgers New Brunswick Honors College

New Brunswick, NJ

B.S. in Computer Science and Data Science — GPA: 3.8

Expected June 2028

- Related Coursework: Data Structures & Algorithms, Computer Architecture, Linear Algebra, Discrete Math

EXPERIENCE

Undergraduate Research Assistant

June 2026 – Present

Rutgers Art and AI Lab

New Brunswick, NJ

- Built Python automation pipelines to process and analyze **80+ images**, converting extracted visual statistics into structured datasets for downstream research analysis
- Visualized and analyzed feature trends across AI-generated and authentic artwork, enabling quantitative evaluation of generative model accuracy and stylistic differences
- Developed reproducible data workflows for image-property aggregation, matrix construction, and comparative statistical analysis in AI image generation research

Web Developer

Jan 2026 – Present

Casual Harmony Acapella

New Brunswick, NJ

- Built and maintained the organization's official website using **React, Vite, and Tailwind CSS**
- Designed polished user interfaces with emphasis on usability and consistent visual branding across devices
- Developed functional contact systems to streamline communication between the organization and prospective members, clients, and audiences

PROJECTS

Rubik's Cube Solver | *Python, PyTorch, Pandas, NumPy*

- Developed a 2x2 Rubik's Cube simulator in **Python** with full move set and scramble generation, producing **50k+ labeled states** to train machine learning models
- Designed and trained a **custom CNN in PyTorch** with residual blocks, dropout, and weight decay, achieving **80%** move-prediction accuracy and full-solve success rate on 5-move scrambles
- Built a **high-throughput data pipeline** using PyTorch Dataset/DataLoader with GPU preprocessing and batching, reducing data prep time and enabling rapid scaling of experiments
- Created evaluation tools for accuracy/loss tracking and benchmarking across repeated trials

Connect 4 | *Java, Swing*

- Built an interactive **Connect Four game in Java** with a Swing GUI, implementing game logic, turn management, and responsive 6x7 gameplay
- Implemented **piece-drop animations**, multithreading, and win/draw detection across all board states, delivering smooth and reliable gameplay
- Tested and refined the game through 50+ full playthroughs and peer feedback, ensuring a polished, **bug-free** user experience

LEADERSHIP

Assistant Business Manager

April 2026 - Present

Casual Harmony Acapella

New Brunswick, NJ

- Organized concerts by coordinating venues, scheduling, logistics, marketing, and cross-team communication
- Directed financial operations by handling budgeting, reimbursements, and fundraising

Board Member

Jan 2026 - Present

Muslim Tech Collaborative

New Brunswick, NJ

- Organized a multi-university hackathon, coordinating logistics for 100+ participants and locating sponsors

TECHNICAL SKILLS

Languages: Java, Python, SQL, JavaScript, HTML/CSS, C/C++

Frameworks/Libraries: React, Node.js, PyTorch, pandas, NumPy, Matplotlib

Developer Tools: Git, GitHub, VS Code, PyCharm, IntelliJ